Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/02/2024
PI Name:	Komatsu, David Ph.D.		
Project Title:	Effects of Simulated Microgravity on Osteoblast-Osteoclast Coupling and Therapeutic Countermeasures		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	David.Komatsu@Stonybrookmedicine.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	631-444-7222
Organization Name:	State University of New York, Stony Brook (Stony Brook University)		
PI Address 1:	Department of Orthopaedics		
PI Address 2:	HSC-T18, Room 085		
PI Web Page:			
City:	Stony Brook	**State:**	NY
Zip Code:	11794-8181	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	02/11/2020	**End Date:**	12/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Ruby, Anna Maria	**Contact Phone:**	321-867-7065
Contact Email:	annamaria.j.ruby@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per NSSC information. (Ed. 12/15/22) NOTE: End date changed to 07/26/2023 per NSSC information. (Ed. 12/15/22) NOTE: End date changed to 07/26/2022 per NSSC information. (Ed. 11/19/21)		

Key Personnel Changes/Previous PI:	NOTE (Dec 2022): Per PI -- Dr. Sardar Uddin's involvement with the project as a formal consultant has ended, though he still helps with discussions (Ed. 12/15/22). NOTE (Dec 2021): Dr. Sardar Uddin remains a Consultant and CoInvestigator on the current award, with Dr. David Komatsu serving as Principal Investigator (PI). The studies are being completed by a pharmacology graduate student, Isaiah Taylor, who joined the laboratory in the spring of 2021. These studies will serve as the core of his Ph.D. dissertation. NOTE (March 2020): New principal investigator is Dr. David Komatsu, who replaces previous PI - Dr. Sardar Uddin; Dr. Uddin remains on the project as a CoInvestigator. Kaitlin Farrell completed her work as a research technician in July of 2023. However, she has been hired with different funds and is still providing some assistance to Isaiah Taylor with the outcomes analyses, as well as overall with lab management.
COI Name (Institution):	Uddin, Sardar Ph.D. (State University of New York, Stony Brook (Stony Brook University))
Grant/Contract No.:	80NSSC18K1479
Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE: Continuation of "Effects of Simulated Microgravity on Osteoblast-Osteoclast Coupling and Therapeutic Countermeasures--PI Uddin" with new Principal Investigator (PI) Dr. David Komatsu, who replaced previous PI, Dr. Sardar Uddin, in February 2020. Objective: Astronauts lose 1-1.5% of their bone mass per month during low Earth orbit missions. Spaceflight profiles of 2 years or longer, as projected for Mars missions, will significantly increase bone fragility, potentially compromising the ability of astronauts to perform their mission tasks in transit and on Mars. To facilitate human exploration of Mars, it is therefore essential to understand the underlying causes of microgravity-induced bone loss and develop effective countermeasures to sustain bone structure and mechanical integrity under microgravity conditions. Our long-term goal is to protect the skeletal health of astronauts in space environments so they can perform their mission operations. The objective of the current proposal is to determine the effects of microgravity on osteoblast-osteoclast coupling and communication in order to identity a countermeasure for microgravity-induced decoupling. Our central hypothesis is that simulated microgravity will impair osteoblast-osteoclast coupling and that a bisphosphonate (alendronate) and/or anti-RANKL antibody (denosumab) treatment will mitigate microgravity-induced osteoblast-osteoclast decoupling. The rationale for this proposal is the fact that microgravity promotes osteoclastic bone resorption and impairs osteoblastic bone formation. It is expected that physiological osteoblast-osteoclast coupling is impaired and that denosumab, a receptor activator of nuclear factor-?B ligand (RANKL) inhibitor, will restore osteoblast-osteoclast coupling and mitigate microgravity-induced bone loss to a greater degree than alendronate, an anti-catabolic agent. Specific Aims and Methods: The central hypothesis will be tested by completing the following two specific aims. 1. Determine the effects of simulated microgravity on osteoblast-osteoclast coupling and communication utilizing in vitro co-cultures. Our working hypothesis is that microgravity will alter osteoblast-osteoclast communication processes. Pathway analyses will be conducted by assessing the differential expression of mRNA (e.g., RNA sequencing, or "RNA-seq"; quantitative polymerase chain reaction, or "qPCR") and protein (i.e., enzyme-linked immunosorbent assay, or "ELISA" tests; Western blots), as well as histology and histomorphometry. Combined Confocal Raman Scanning Electron Microscopy (CRSEM) will be used to determine functional impairment of bone resorption and formation. 2. Assess the efficacy of denosumab relative to alendronate as a countermeasure to microgravity-induced osteoblast-osteoclast decoupling in vitro. Our working hypothesis is that denosumab will be more efficacious in restoring microgravity-induced impairment of osteoblast-osteoclast coupling as it functions by blocking excess RANKL, while alendronate only affects resorption. Osteoblast-osteoclast communication will be assessed using proteomic and genetic analysis and function restoration will be assessed using histology, histomorphometry, and CRSEM. Significance and Outcomes: Bone remodeling is a complex process that is tightly regulated by osteoblast-osteoclast coupling and communication. Microgravity impairs this process by increasing osteoclast-associated resorption and decreasing osteoblastic bone formation. In order to continue to expand our crewed spaceflight program, it is essential to determine the effects of microgravity on osteoblast-osteoclast coupling and develop targeted countermeasures to sustain bone health in microgravity. At the completion of the study, we expect to characterize the effects of microgravity on osteoblast-osteoclast coupling, and identify communication pathways associated with microgravity, as well as establish the efficacy of alendronate and/or denosumab as potential therapeutics to restore osteoblast-osteoclast communication and function. The outcomes of this study will fill NASA's recognized knowledge gap and guiding question in microgravity effects on osteoblast-osteoclast communication and evaluate potential therapeutics to the aberrant microgravity-induced bone loss in astronauts (AH2, 3, CMB1, 2, 6). [Ed. note Dec 2021: NASA research gaps have changed since original proposal. For information on Animal Biology (formerly AH), or Cell and Molecular Biology (CMB), see Space Biology program: https://]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research is highly innovative as it uses, for the first time, creative osteoblast-osteoblast communication on the bone chip to study bone formation and resorption. The study will use innovative experimental approaches to study bone formation and bone resorption, resulting in the first comprehensive in vitro hylomorphic analysis of bone formation and resorption relative to changing gravity and osteoblast-osteoclast activity. The successful completion of research will allow us to develop new approaches to studies of osteoblast-osteoclast communication in different gravitational and pathological conditions. Studying the changes in osteoblast-osteoclast coupling will enable us to identify new therapeutic targets to facilitate treatment of bone loss in space and Earth based conditions.

Task Progress:	December 2024 Update: Final Reporting per the Principal Investigator (PI) [Ed., 12/2/24.] Since the last reporting period, studies have been repeated with primary osteoblasts and osteoclasts under simulated microgravity (SMG). We have finalized experiments in the following areas: CELL GROWTH • Scanning Electron Microscopy (SEM): In order to ensure that cells would successfully grow on decellularized human tricortical blocks, we seeded MC3T3 cells, a murine calvarial osteoblastic cell line, on blocks and cultured them for up to 7 days under normal gravity. The blocks were then fixed, dehydrated, and sputter-coated with gold and imaged with SEM. • Live Dead: To further assess cell growth on the tricortical blocks, we next seeded MC3T3 cells on tricortical blocks and cultured them for up to 28 days under normal gravity. The blocks were then stained en bloc using a live dead kit (Sigma), which stains live cells green and dead cells red. The blocks were then imaged using confocal microscopy. BIOMECHANICS After establishing proper seeding conditions and ensuring cell growth on the tricortical blocks, we initated experiments with primary human cells. To determine the effects of simulated microgravity on biomechanical properties, we cultured primary human osteoblasts or primary human macrophages differentiated into osteoclasts on tricortical human decellularized blocks for 28 days under either normal gravity or simulated microgravity (MG). The biomechanical integrity of tricortical blocks was assessed using unconstrained uniaxial compression. Briefly, the blocks were placed between two stainless steel plates and a monotonic load to failure was applied at 5mm/min until failure using a compression tester (MTEST Quatro, Admet). Force and displacement were digitally sampled at 100 Hz and exported to Excel (Microsoft) for analysis. Parameters analyzed included stiffness, energy to failure, yield force, ultimate force, and failure force. The integrity of blocks seeded with osteoblasts was significantly reduced in MG-exposed samples compared to samples experiencing normal gravity. FUTURE DIRECTIONS In addition to these finalized experiments, we have collected samples for additional analyses that are still ongoing. This includes weekly media collected from normal and microgravity cultures of osteoblasts and osteoclasts over 28 days which will be used for enzyme-linked immunosorbent assay (ELISA) analysis of signaling molecules. As well, tricortical blocks from normal and microgravity cultures of osteoblasts and osteoclasts incubated for 28 days have been collected for gene expression analyses. Finally, tricortical blocks from normal and microgravity cultures of osteoblasts and osteoclasts incubated for 28 days have been fixed and embedded in PMMA for histomorphometric analyses. We intend to complete these studies within the next year and publish a manuscript with the combined results.
Bibliography Type:	Description: (Last Updated: 07/03/2024)